

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

01st December 2020

**Subject SP7 – General Insurance Reserving and
Capital Modelling**

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Mark allocations are shown in brackets.***

- Q. 1) i)** A junior actuarial analyst was asked to build a reserving triangle based on incurred claims reported till date. The data is available for the last 4 years. He is confused about using quarterly or annual development cohorts and has sought your suggestion. List the points you would ask him to consider when deciding between quarterly or annual triangulations in reserving. (7)
- ii)** You are an actuary working for a GI company writing Motor & Liability insurance. State the circumstances under which you would select each of the two methods below for quantifying the level of uncertainty in reserve estimates.
- The Mack method
 - Bootstrapping the over dispersed Poisson
- (4)
- iii)** The company had started underwriting 3 year Specialty policies from 2014 and wants to estimate the profitability of the business written so far for management reporting.

Following premium and claims details have been provided (amounts in Rs ‘000s):

Year	GWP
2014	60,000
2015	75,000
2016	1,00,000
2017	1,10,000
2018	90,000
2019	1,20,000
2020	1,25,000

Policy sold	Claim events		Claims Reported		Payment information	
Year	Year	Claim estimate	Year	Amount claimed	Year	Actual Claim cost
2014	2014	10,000	2014	10,000	2015	10,000
2014	2015	15,000	2015	15,000	2016	20,000
2014	2017	15,000	2017	12,000	2017	15,000
2015	2015	15,000	2015	15,000	2016	15,000
2015	2017	1,00,000	2018	1,00,000	2018	1,00,000
2015	2016	15,000	2016	15,000	2017	15,000
2016	2017	10,000	2018	12,000	2020	14,500
2016	2019	20,000	2019	20,000	2019	25,000
2016	2016	10,000	2017	15,000	2017	20,000
2016	2017	10,000	2017	10,000	2018	15,000
2017	2018	15,000	2019	15,000	2020	20,000
2017	2017	20,000	2017	20,000	2017	20,000
2017	2018	12,000	2019	14,000	2019	15,000
2017	2017	10,000	2018	10,000	2018	10,000
2018	2018	25,000	2018	25,000	2019	15,000
2018	2019	25,000	2019	25,000	2020	25,000
2019	2019	20,000	2020	20,000	2020	20,000
2019	2020	20,000	2020	20,000	2020	20,000
2020	2020	2,000	2020	2,500	2020	NA

- a) Estimate the appropriate loss ratios for policies sold in each of the respective years. Ignore all other incomes and expenses. (11)
 - b) Mention any assumptions and rationale for your decisions for calculating the above estimates. (2)
 - c) The initial Loss Ratio expected over the term of the policy was at 75%. Comment on the appropriateness of the ILR. (2)
- [26]**

- Q. 2)**
- i) A large general insurer has decided to adopt stochastic modelling to calculate its capital requirements. Explain the possible reasons (with help of examples) for the insurer to choose stochastic modelling approach instead of simpler deterministic approach. (5)
 - ii) The company has collected all the data required for capital modelling. Discuss the approach the company should follow to model its Gross Underwriting Claim Risk. Your answer should also discuss the class structure, distribution assumptions and correlations. (15)
 - iii) The company has recently been approached by TechGRO, a firm seeking insurance cover for its business involving tech-based delivery of grocery and household items/appliances. State, with reasons, the types of insurance cover TechGRO might want to purchase. (8)
- [28]**

- Q. 3)**
- i) An Investor analyst was looking at the combined ratio (COR) of a GI company to measure the profitability of a class of business. The analyst was advised to also consider ROE as an alternative measure of success and not rely solely on COR indicator for assessing profitability. Suggest the reason for such advice. (4)
 - ii) From the following financial information of two GI companies A and B as at 31st March 2020, stating any assumptions made.
 - a) Construct the balance sheet as at 31st March 2020. *(Note: Use of table is not required. Kindly write the entries below the appropriate header)*
 eg. Company A B
 xyz xx xx) (5)
 - b) Derive and comment on the underwriting, solvency and return on capital employed ratios for these two companies. Taxation should be ignored. (12)
 - c) Define two other insurance related ratios that could be derived from the data to compare the investment performance and profitability of unexpired risk of these two companies. (4)

Company (Amounts in Rs crores)	A	B
Gross Written Premium	5.0	200
Additional Unexpired Risk Reserve c/f	1.8	15
Gross Outstanding Claims Reserve b/f	2.0	80
Gross Claims Paid	3.5	70

Gross Outstanding Claims Reserve c/f	3.0	85
Non Acquisition Expenses	0.50	25
Investment Income	0.30	10
Current Assets at 31/03/2020	0.50	8
Current Liabilities at 31/03/2020	1.1	10
Investments at year end	12.8	355
Share capital at 31/03/2020	1.5	50
Acquisition Costs as a % of Gross Written Premium	30%	15%

[25]

- Q. 4)**
- i)** Discuss the issues which make reserving for a reinsurance company more difficult than for a primary insurance company? (5)
 - ii)** Briefly describe components of a Reinsurer's loss reserve. (5)
 - iii)** Outline the steps you will perform while calculating reserves for a reinsurer. (4)
 - iv)** A student actuary has been asked to carry out reserve analysis for the following categories of reinsurance. Outline the approach that the student actuary should consider.
 - a)** Treaty Casualty excess of loss (2)
 - b)** Treaty property catastrophe (2)
 - v)** An insurance company wants to purchase Stop Loss reinsurance for its Motor Third Party liability portfolio. It approaches three reinsurance companies for cover but they decline. Explain why reinsurers might be reluctant to provide Stop Loss cover. (3)
- [21]**
